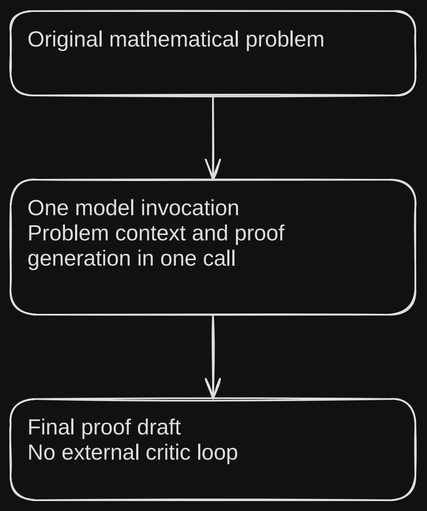
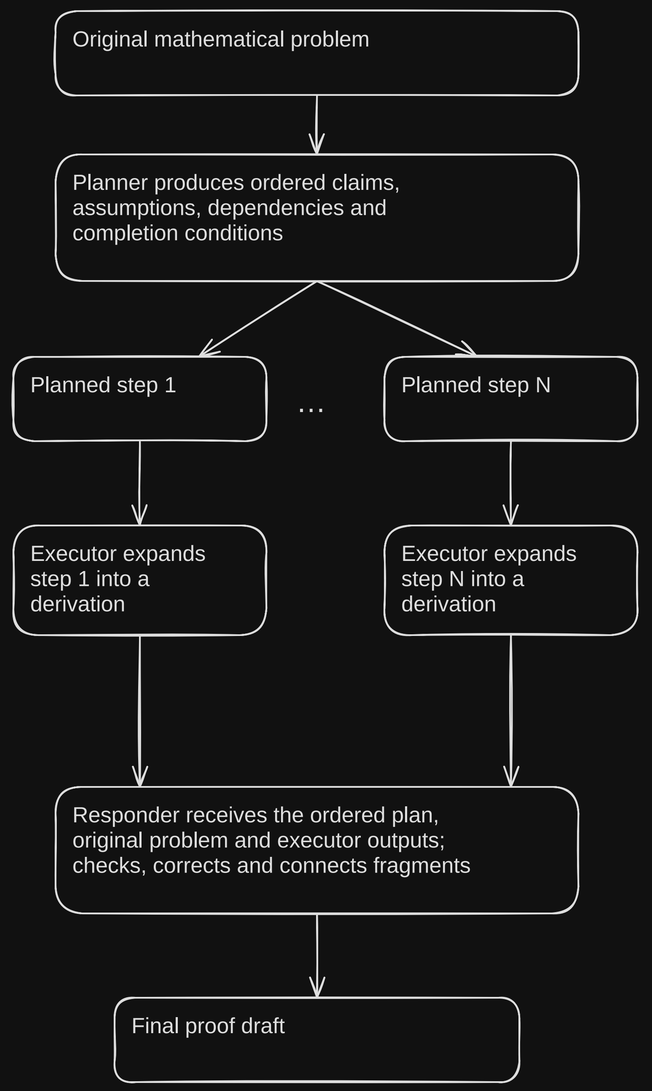
Same language model. Three ways to manage mathematical context.

Case study: “Prove the Prime Number Theorem” | IFM/K2-Horizon-375B-A23B

1. Standalone model

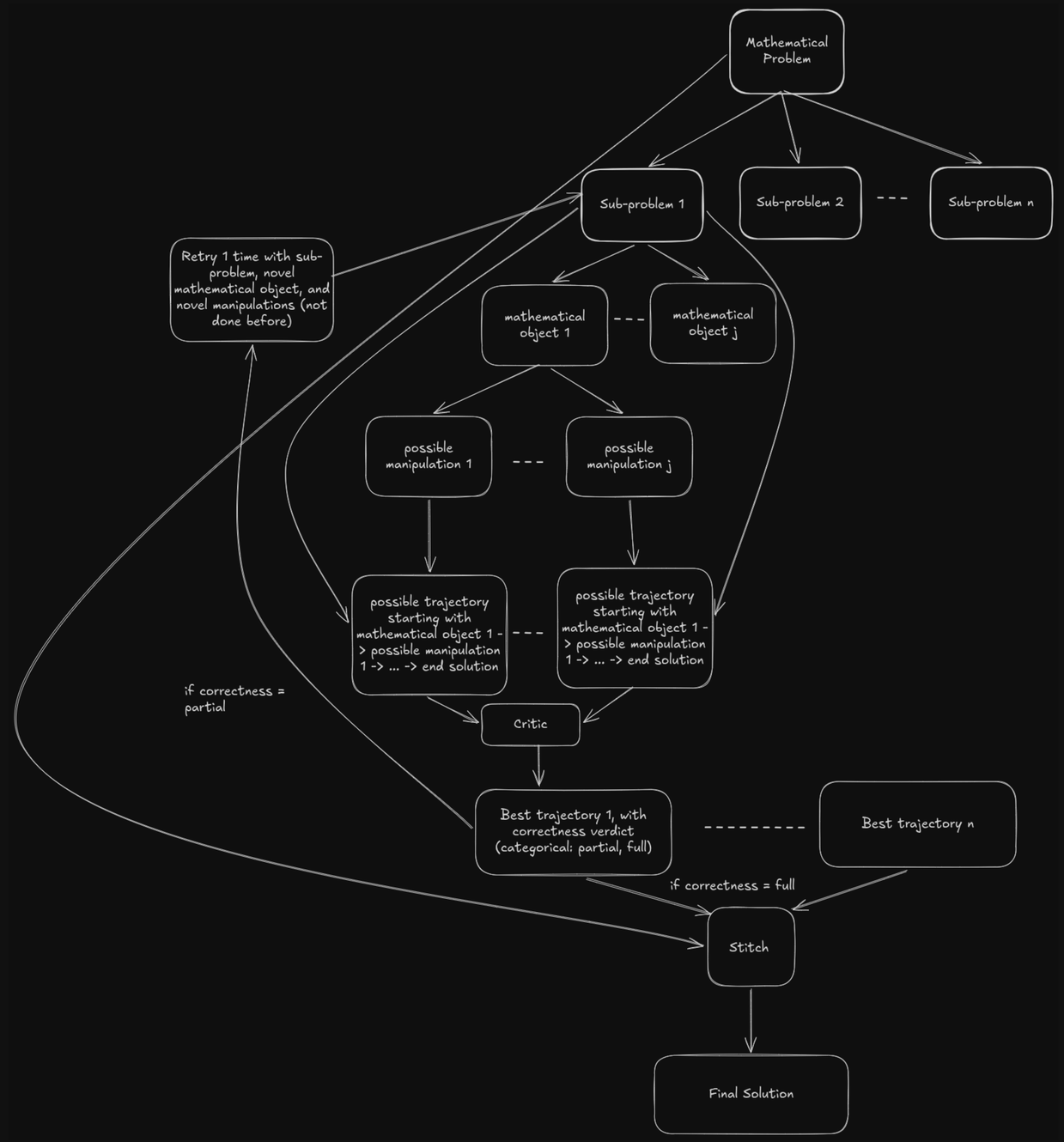


2. Planner–executor



No automatic replanning or critic loop.
Blocked dependencies remain partial.
Plan and outputs are saved in run.json.
N is the number of planned steps.
Sequential execution in dependency order.
Each call receives the original problem, local claim and required predecessor proofs only.

3. Math Harness



Supplied source figure retained above. Current code: subproblems 1…i; objects 1…j; manipulations 1…k. All calls are sequential. Stitching also accepts partial or unranked work; a full verdict is not required.

OBSERVED OUTPUTS — qualitative review of one draft per approach

Standalone: invalid asymptotic step.
The PNT conclusion was not established.

Planner–executor: contour and zero-sum development; major analytic inputs and local gaps remain.

Math Harness: broader supporting derivations; local errors and an unproved Tauberian lemma.

Unequal inference budgets; incomplete multi-stage work. Qualitative comparison, not a controlled benchmark or formal verification.

github.com/antibloch/math_harness